Triangles & Congruence

ClipSAT Printable Practice Suite (clipsat.org) • Topic 3.01

Part I — Free-Response Questions (FRQs): Foundational Fluency (1–7)

1 [Foundational] • Target: Review: Equations of lines

Find the positive solutions to the quadratic equation $x^2 - 6x + 5 = 0$.

2 [Foundational] • Target: Review: Equations of lines

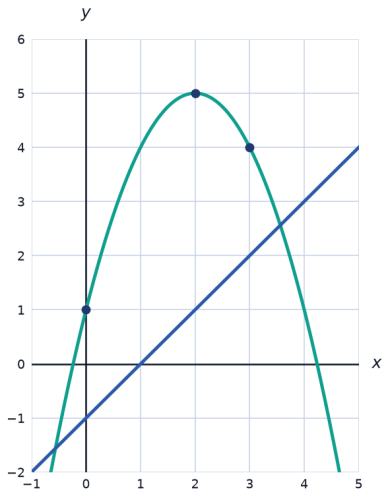
A data set recorded for review: equations of lines consists of the four values $\{15, 19, 23, 29\}$. What is the arithmetic mean? Show your complete algebraic work.

3 [Foundational] • Target: Review: Equations of lines

For the model $f(x) = 6x + 3$ in review: equations of lines, evaluate $f(3)$.

4 [Foundational] • Target: Review: Equations of lines

Find the positive solutions to the quadratic equation $x^2 - 8x + 12 = 0$.



5 [Foundational] • Target: Review: Equations of lines

For the model $f(x) = 6x + 3$ in review: equations of lines, evaluate $f(4)$.

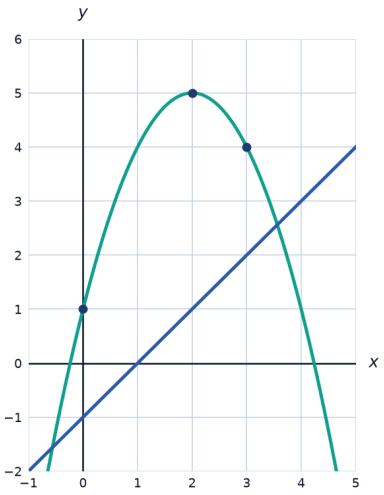
6 [Foundational] • Target: Review: Equations of lines

Find the positive solutions to the quadratic equation $x^2 - 10x + 21 = 0$.

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7 [Foundational] • Target: Review: Equations of lines

For the model $f(x) = 6x + 3$ in review: equations of lines, evaluate $f(5)$.



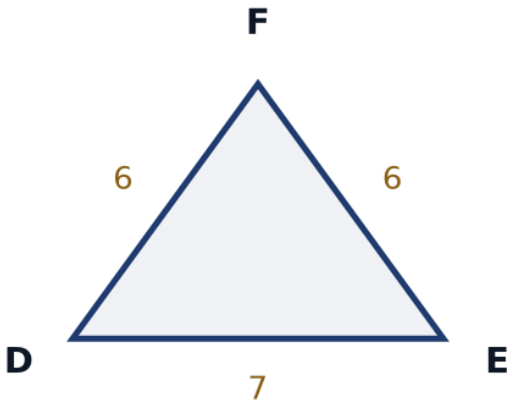
Part II — Free-Response Questions (FRQs): Intermediate Multi-Step Application (8–14)

8 [Intermediate] • Target: Review: Equations of lines

Find the positive solutions to the quadratic equation $x^2 - 12x + 32 = 0$.

9 [Intermediate] • Target: Equations of parallel and perpendicular lines

In ΔABC studying equations of parallel and perpendicular lines, $m\angle A = 45^\circ$ and $m\angle B = 45^\circ$. What is $m\angle C$? Show your complete algebraic work.



10 [Intermediate] • Target: Equations of parallel and perpendicular lines

A geometric figure scaled by factor $k = 2$ has an original side length of 4 cm. What is the corresponding image length? Show your complete algebraic work.

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11 [Intermediate] • Target: Equations of parallel and perpendicular lines

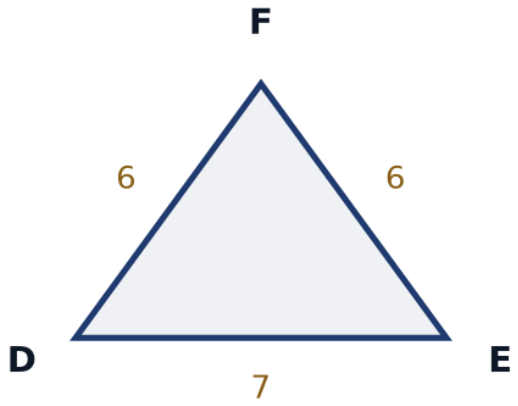
On the coordinate plane, a line with slope $m = 5$ passes through $(0, 4)$. What is its y-coordinate when $x = 1$? Show your complete algebraic work.

12 [Intermediate] • Target: Equations of parallel and perpendicular lines

In ΔABC studying equations of parallel and perpendicular lines, $m\angle A = 45^\circ$ and $m\angle B = 52^\circ$. What is $m\angle C$? Show your complete algebraic work.

13 [Intermediate] • Target: Equations of parallel and perpendicular lines

A geometric figure scaled by factor $k = 3$ has an original side length of 4 cm. What is the corresponding image length? Show your complete algebraic work.



14 [Intermediate] • Target: Equations of parallel and perpendicular lines

On the coordinate plane, a line with slope $m = 5$ passes through $(0, 4)$. What is its y-coordinate when $x = 2$? Show your complete algebraic work.

Part III — Free-Response Questions (FRQs): Challenging & Cross-Lesson Synthesis (15–20)

15 [Challenging] • Target: Equations of parallel and perpendicular lines • Cross-Lesson: 8.01 Area, Surface Area & Volume

In ΔABC studying equations of parallel and perpendicular lines, $m\angle A = 45^\circ$ and $m\angle B = 59^\circ$. What is $m\angle C$? Show your complete algebraic work.

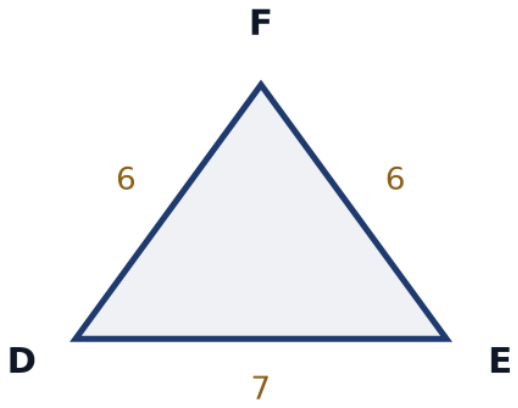
16 [Challenging] • Target: Equations of parallel and perpendicular lines • Cross-Lesson: 9.01 Transformations & Symmetry

A geometric figure scaled by factor $k = 4$ has an original side length of 4 cm. What is the corresponding image length? Show your complete algebraic work.

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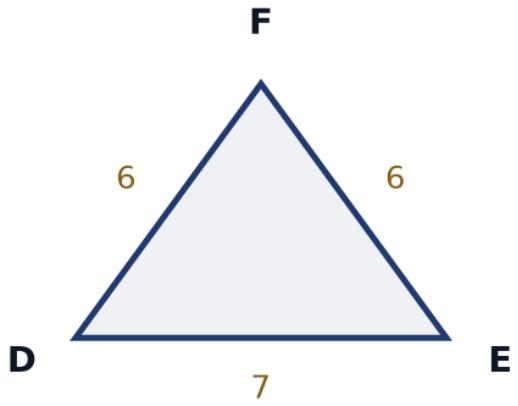
17 [Challenging] • Target: Equations of parallel and perpendicular lines • Cross-Lesson: 1.01 Foundations & Reasoning

On the coordinate plane, a line with slope $m = 5$ passes through $(0, 4)$. What is its y-coordinate when $x = 3$? Show your complete algebraic work.



18 [Challenging] • Target: Equations of parallel and perpendicular lines • Cross-Lesson: 10.01 Coordinate Geometry

In ΔABC studying equations of parallel and perpendicular lines, $m\angle A = 45^\circ$ and $m\angle B = 66^\circ$. What is $m\angle C$? Show your complete algebraic work.



19 [Challenging] • Target: Equations of parallel and perpendicular lines • Cross-Lesson: 11.01 Probability

A geometric figure scaled by factor $k = 5$ has an original side length of $4\ \text{cm}$. What is the corresponding image length? Show your complete algebraic work.

20 [Challenging] • Target: Equations of parallel and perpendicular lines • Cross-Lesson: 2.01 Lines & Angles

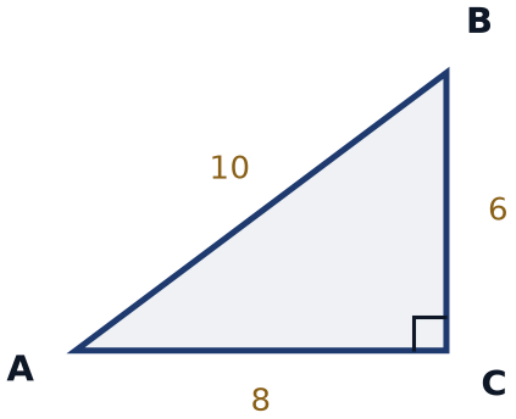
On the coordinate plane, a line with slope $m = 5$ passes through $(0, 4)$. What is its y-coordinate when $x = 4$? Show your complete algebraic work.

Part IV — Multiple-Choice Questions (MCQs): Foundational Concept Mastery (21–28)

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21 [Foundational] • Target: Parallel lines and transversals

In ΔABC studying parallel lines and transversals, $m\angle A = 42^\circ$ and $m\angle B = 45^\circ$. What is $m\angle C$?



a. 93°

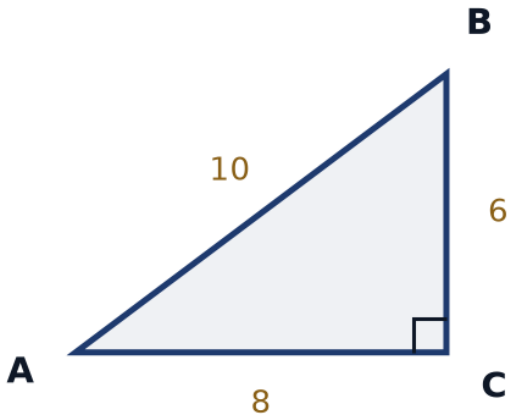
b. 103°

c. 85°

d. 138°

22 [Foundational] • Target: Parallel lines and transversals

A geometric figure scaled by factor $k = 3$ has an original side length of 6 cm. What is the corresponding image length?



a. 15 cm

b. 18 cm

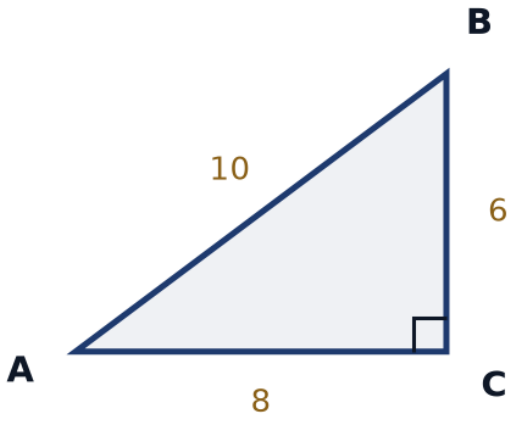
c. 24 cm

d. 54 cm

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23 [Foundational] • Target: Parallel lines and transversals

On the coordinate plane, a line with slope $m = 2$ passes through $(0, 8)$. What is its y-coordinate when $x = 3$?



a. 12

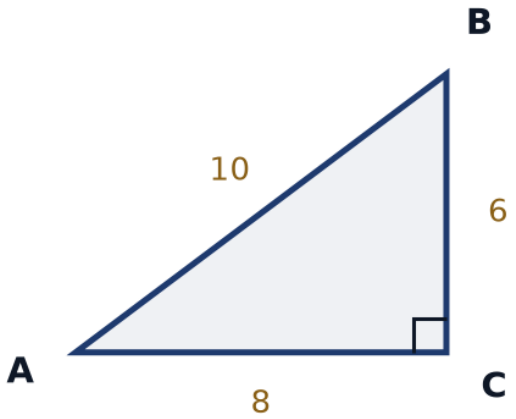
b. 16

c. 14

d. 16

24 [Foundational] • Target: Parallel lines and transversals

In ΔABC studying parallel lines and transversals, $m\angle A = 42^\circ$ and $m\angle B = 66^\circ$. What is $m\angle C$?



a. 72°

b. 82°

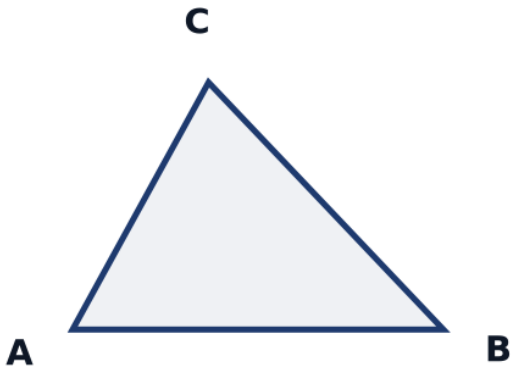
c. 64°

d. 138°

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25 [Foundational] • Target: Proving lines parallel

In ΔABC studying proving lines parallel, $m\angle A = 43^\circ$ and $m\angle B = 45^\circ$. What is $m\angle C$?



a. 92°

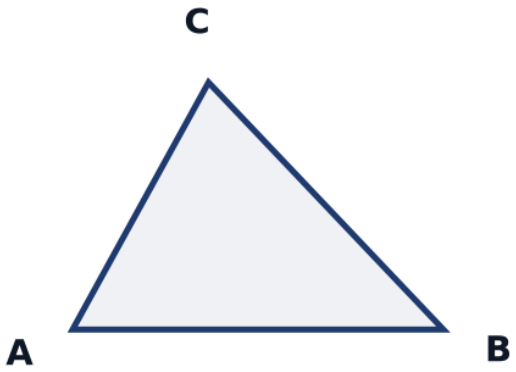
b. 102°

c. 84°

d. 137°

26 [Foundational] • Target: Proving lines parallel

A geometric figure scaled by factor $k = 3$ has an original side length of 7 cm. What is the corresponding image length?



a. 18 cm

b. 21 cm

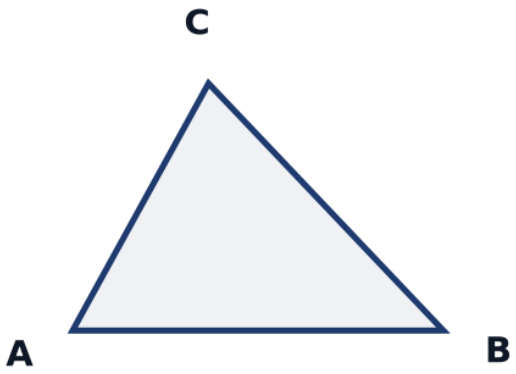
c. 28 cm

d. 63 cm

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27 [Foundational] • Target: Proving lines parallel

On the coordinate plane, a line with slope $m = 3$ passes through $(0, 9)$. What is its y-coordinate when $x = 3$?



a. 15

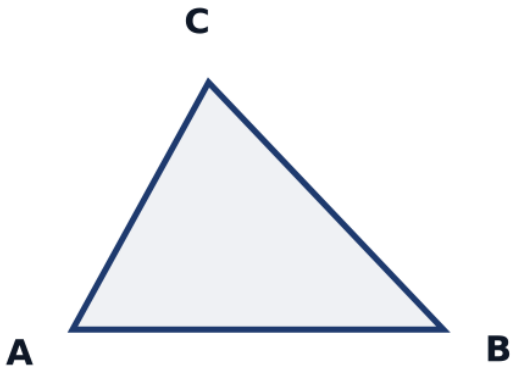
b. 20

c. 18

d. 21

28 [Foundational] • Target: Proving lines parallel

In ΔABC studying proving lines parallel, $m\angle A = 43^\circ$ and $m\angle B = 66^\circ$. What is $m\angle C$?



a. 71°

b. 81°

c. 63°

d. 137°

Part V — Multiple-Choice Questions (MCQs): Intermediate Exam Application (29–37)

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29 [Intermediate] • Target: Parallel lines and transversals

A geometric figure scaled by factor $k = 2$ has an original side length of 6 cm. What is the corresponding image length?

a. 10 cm

b. 12 cm

c. 18 cm

d. 24 cm

30 [Intermediate] • Target: Parallel lines and transversals

On the coordinate plane, a line with slope $m = 2$ passes through $(0, 8)$. What is its y-coordinate when $x = 1$?

a. 8

b. 12

c. 10

d. 12

31 [Intermediate] • Target: Parallel lines and transversals

On the coordinate plane, a line with slope $m = 2$ passes through $(0, 8)$. What is its y-coordinate when $x = 2$?

a. 10

b. 14

c. 12

d. 14

32 [Intermediate] • Target: Parallel lines and transversals

A geometric figure scaled by factor $k = 4$ has an original side length of 6 cm. What is the corresponding image length?

a. 20 cm

b. 24 cm

c. 30 cm

d. 96 cm

33 [Intermediate] • Target: Parallel lines and transversals

On the coordinate plane, a line with slope $m = 2$ passes through $(0, 8)$. What is its y-coordinate when $x = 4$?

a. 14

b. 18

c. 16

d. 18

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34 [Intermediate] • Target: Proving lines parallel

On the coordinate plane, a line with slope $m = 3$ passes through $(0, 9)$. What is its y-coordinate when $x = 1$?

a. 9

b. 14

c. 12

d. 15

35 [Intermediate] • Target: Proving lines parallel

In ΔABC studying proving lines parallel, $m\angle A = 43^\circ$ and $m\angle B = 52^\circ$. What is $m\angle C$?

a. 85°

b. 95°

c. 77°

d. 137°

36 [Intermediate] • Target: Proving lines parallel

In ΔABC studying proving lines parallel, $m\angle A = 43^\circ$ and $m\angle B = 59^\circ$. What is $m\angle C$?

a. 78°

b. 88°

c. 70°

d. 137°

37 [Intermediate] • Target: Proving lines parallel

A geometric figure scaled by factor $k = 5$ has an original side length of 7 cm. What is the corresponding image length?

a. 30 cm

b. 35 cm

c. 42 cm

d. 175 cm

Part VI — Multiple-Choice Questions (MCQs): Challenging & Cross-Lesson Synthesis (38–45)

38 [Challenging] • Target: Review: Equations of lines • Cross-Lesson: 10.01 Coordinate Geometry

A data set recorded for review: equations of lines consists of the four values $\{15, 19, 23, 29\}$. What is the arithmetic mean?

a. 19.5

b. 20.5

c. 21.5

d. 23.5

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39 [Challenging] • Target: Equations of parallel and perpendicular lines • Cross-Lesson: 11.01 Probability

On the coordinate plane, a line with slope $m = 5$ passes through $(0, 4)$. What is its y-coordinate when $x = 1$?

a. 4

b. 11

c. 9

d. 14

40 [Challenging] • Target: Equations of parallel and perpendicular lines • Cross-Lesson: 2.01 Lines & Angles

On the coordinate plane, a line with slope $m = 5$ passes through $(0, 4)$. What is its y-coordinate when $x = 2$?

a. 9

b. 16

c. 14

d. 19

41 [Challenging] • Target: Equations of parallel and perpendicular lines • Cross-Lesson: 4.01 Similarity & Proportional Reasoning

In ΔABC studying equations of parallel and perpendicular lines, $m\angle A = 45^\circ$ and $m\angle B = 59^\circ$. What is $m\angle C$?

a. 76°

b. 86°

c. 68°

d. 135°

42 [Challenging] • Target: Equations of parallel and perpendicular lines • Cross-Lesson: 5.01 Right Triangles & Trigonometry

A geometric figure scaled by factor $k = 5$ has an original side length of 4 cm. What is the corresponding image length?

a. 15 cm

b. 20 cm

c. 24 cm

d. 100 cm

43 [Challenging] • Target: Extension: Perpendiculars and distance • Cross-Lesson: 6.01 Polygons & Quadrilaterals

Find the positive solutions to the quadratic equation $x^2 - 4x + 3 = 0$.

a. $x = 1$ and $x = 4$

b. $x = 1$ and $x = 3$

c. $x = -1$ and $x = -3$

d. $x = 4$ and $x = 3$

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44 [Challenging] • Target: Extension: Perpendiculars and distance • Cross-Lesson: 7.01 Circles

For the model $f(x) = 3x + 5$ in extension: perpendiculars and distance, evaluate $f(3)$.

a. 14

b. 17

c. 9

d. 18

45 [Challenging] • Target: Extension: Perpendiculars and distance • Cross-Lesson: 8.01 Area, Surface Area & Volume

Find the positive solutions to the quadratic equation $x^2 - 8x + 15 = 0$.

a. $x = 3$ and $x = 6$

b. $x = 3$ and $x = 5$

c. $x = -3$ and $x = -5$

d. $x = 8$ and $x = 15$

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